

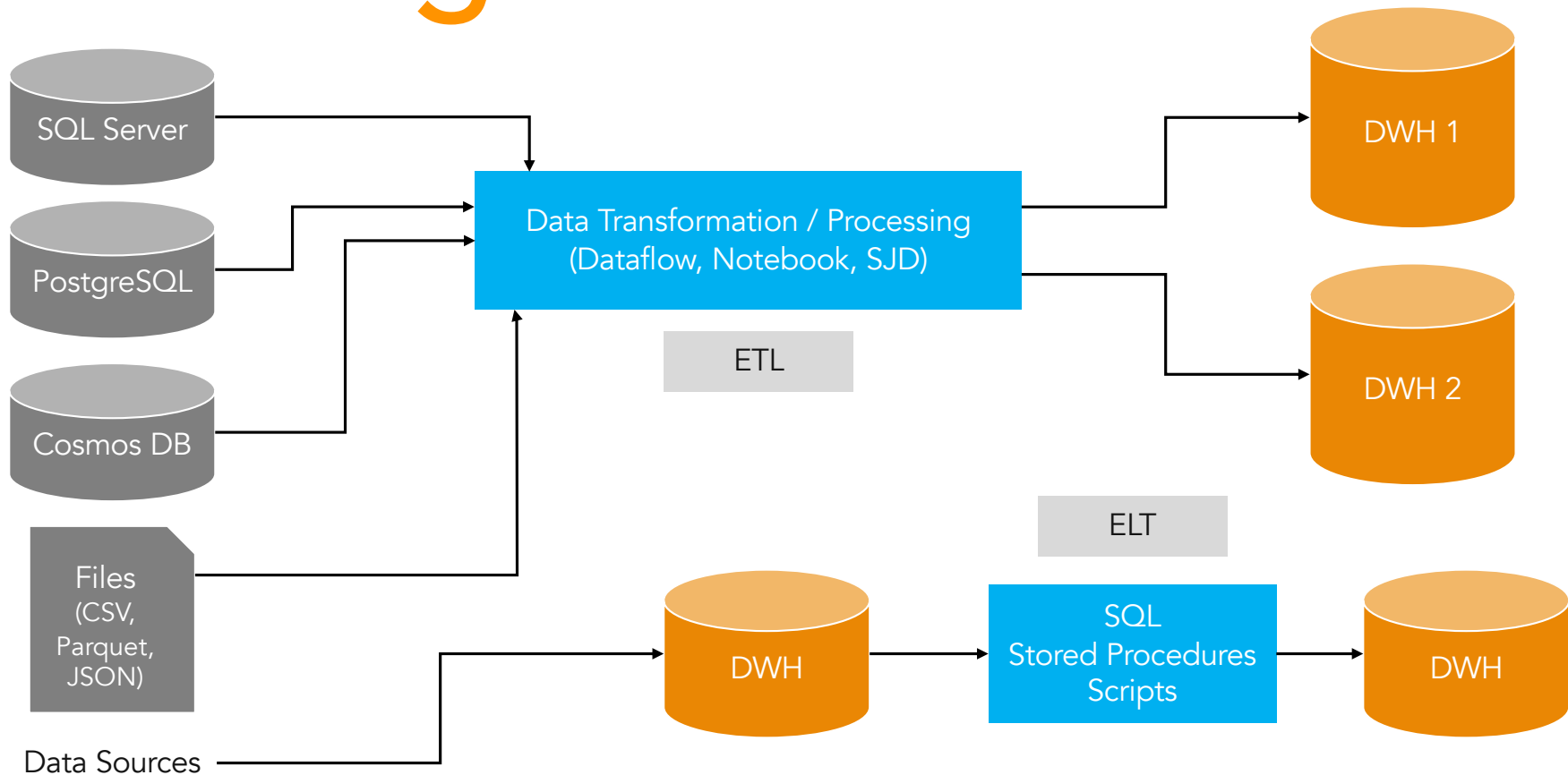
Fabric Data Warehouse

www.3aayaam.in

What is Data Warehouse (DWH)

- Online **Analytical** Processing (OLAP).
- Used mostly for reporting & BI queries. Can be used to run complex ad-hoc queries also.
- Execute complex SQL statements.
 - Lakehouse is not a good fit to run very complex SQL statements.
- RDBMS engine.
- Data is organized around business use cases that are used in reporting queries.
- Used to store Historical Data. Few months to few years.
- ETL is used to load data from OLTP sources to DWH targets. Historical data is stored.
- Data is not modified as much. More than 90% of the queries are SELECT statements.
- Used in GOLD layer of Medallion Architecture.
- **Historical Analytics. Predictive Analytics.**

Loading DWH

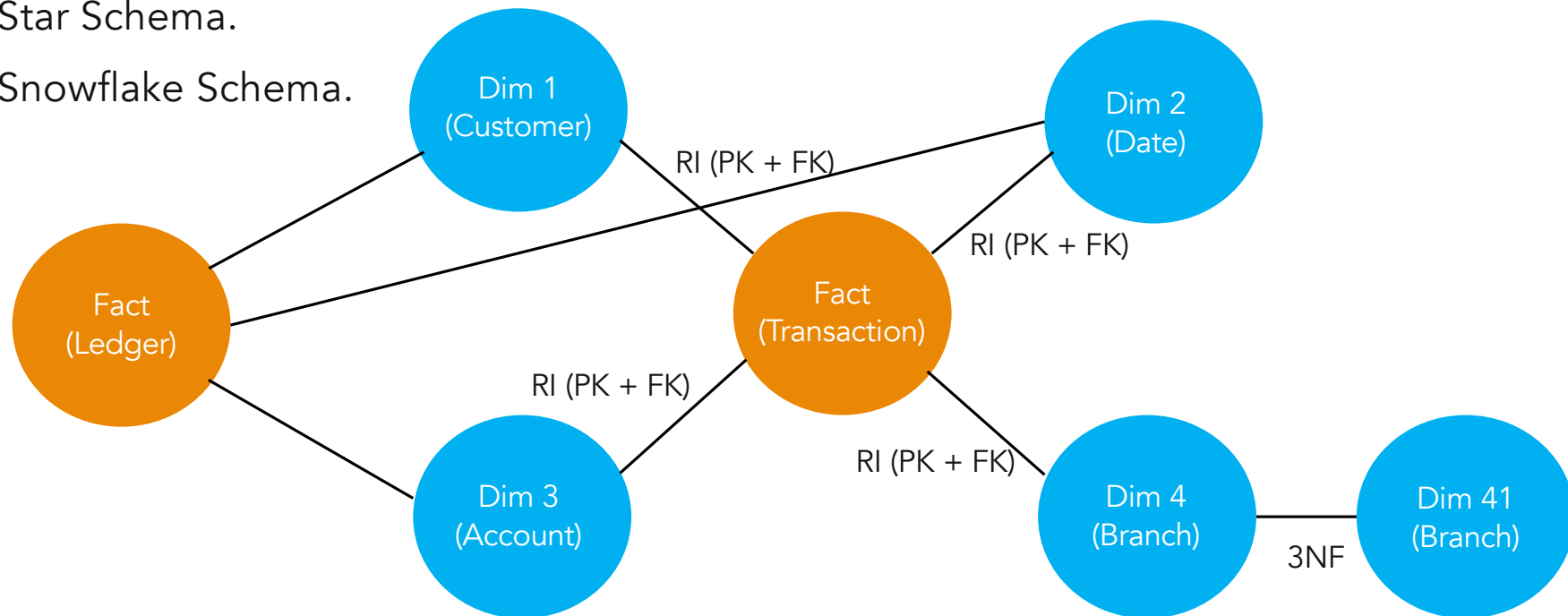


Dimensional Modeling

- Data modeling construct to store data in Data Warehouse systems.
- Purpose of this data model is to focus on read queries and their performance.
- Types of tables: Fact tables & Dimension tables.
- Fact Tables:
 - Used to store business process.
 - Contains data that can be quantified. E.g. price, count, quantity, discount, acc balance etc. Transaction data.
 - Data are added or modified frequently.
- Dimension Table
 - Used to store (mostly) static data.
 - Data that do not change often. E.g. Customer, Account, Date, Product.
- Data models are **entirely based on Reporting or Business Intelligence** requirements.

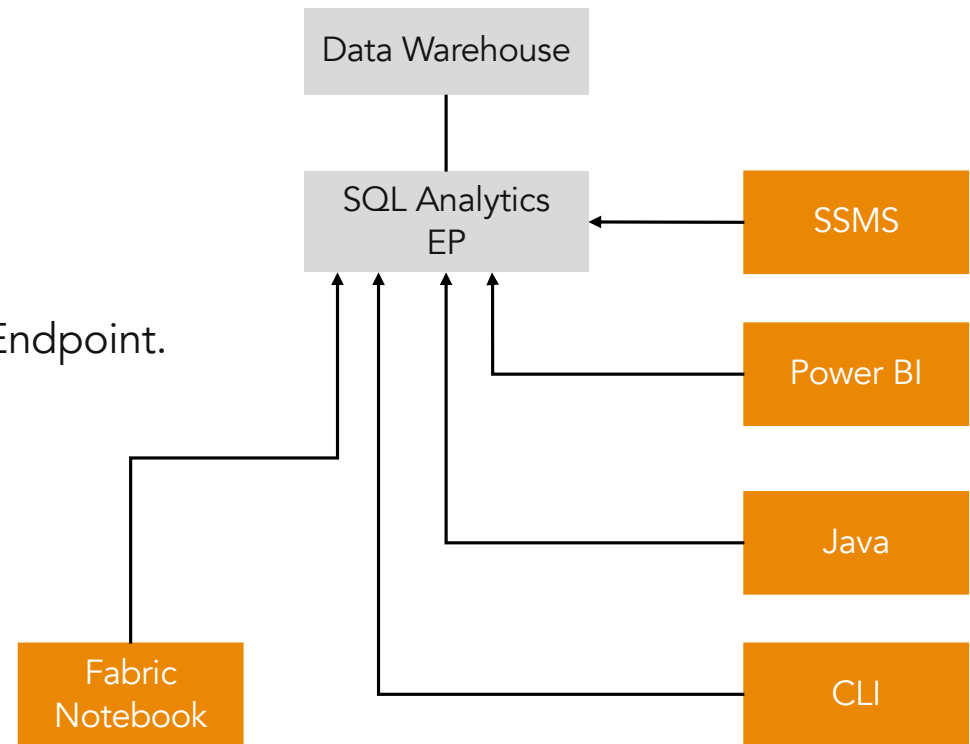
Dimensional Modeling Schemas

- Star Schema.
- Snowflake Schema.



Fabric Warehouse Workload

- Fabric Data Warehouse.
- SQL Analytics Endpoint.
- Connecting to DWH:
 - From console
 - Using connection string of SQL Analytics Endpoint.
 - SSMS (SQL Server Management Studio).
 - Power BI.
 - JDBC & ODBC driver.
 - Fabric Python Notebook.
- Explore Fabric Warehouse UI.



CI/CD Integration

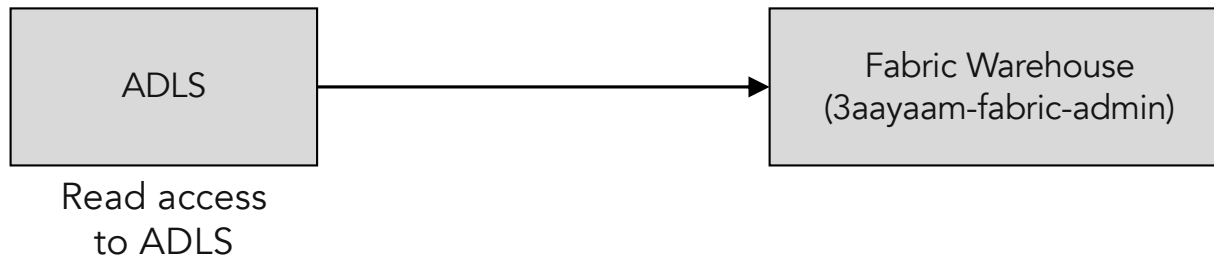
Mode	Tools and features
Fabric web (no Git)	Live development, no version control
Fabric web (with Git)	Live development + version control + continuous integration with workspace branching
IDE / local development	Visual Studio Code with DacFx for schema management and scripting, SSMS for interactive development
Fabric deployment pipelines	Automated deployment, Deployment tracking, Stage promotion
External CI/CD	Azure DevOps, GitHub, REST APIs for automation

Ingest Data

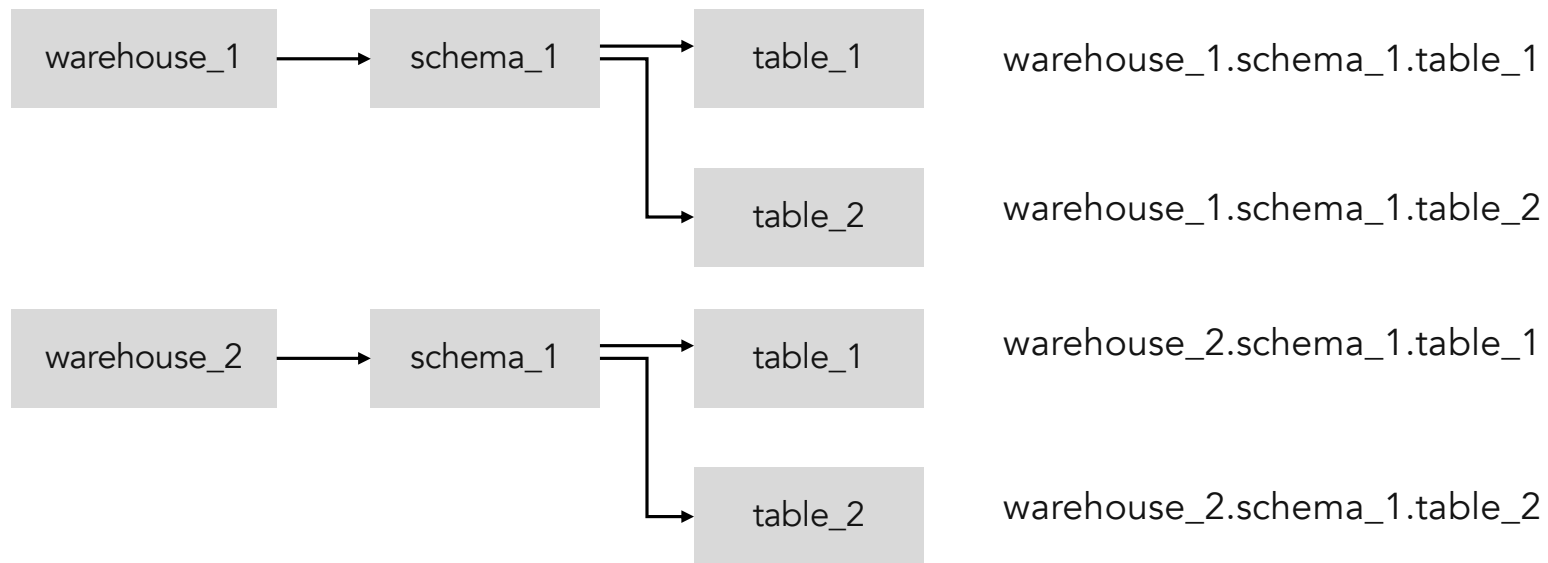
- Copy Job. Dataflow Activity.
- (Using T-SQL) COPY INTO command – Best ingestion throughput. ADLS Gen2 source.
 - Create table
 - `COPY INTO <warehouse>.<schema>.<table> FROM <path> FILE_TYPE = 'CSV' 'PARQUET'`
 - For CSV : FIELDTERMINATOR, ROWTERMINATOR, FIRSTROW = 2, MATCH_COLUMN_COUNT.
 - ERRORFILE, CREDENTIAL, ERRORFILE_CREDENTIAL.
- T-SQL CTAS:
 - CREATE TABLE AS SELECT ...
 - INSERT INTO ... SELECT ...
 - SELECT INTO ...
- DBT.

COPY INTO

```
COPY INTO <schema>.<table>
FROM 'ADLS location'
WITH
(
  FILE_TYPE = 'CSV' | 'Parquet',
  ERRORFILE = 'dir where rejected rows would be written',
  FIRSTROW = 2,
  DATEFORMAT = 'date format'
)
```

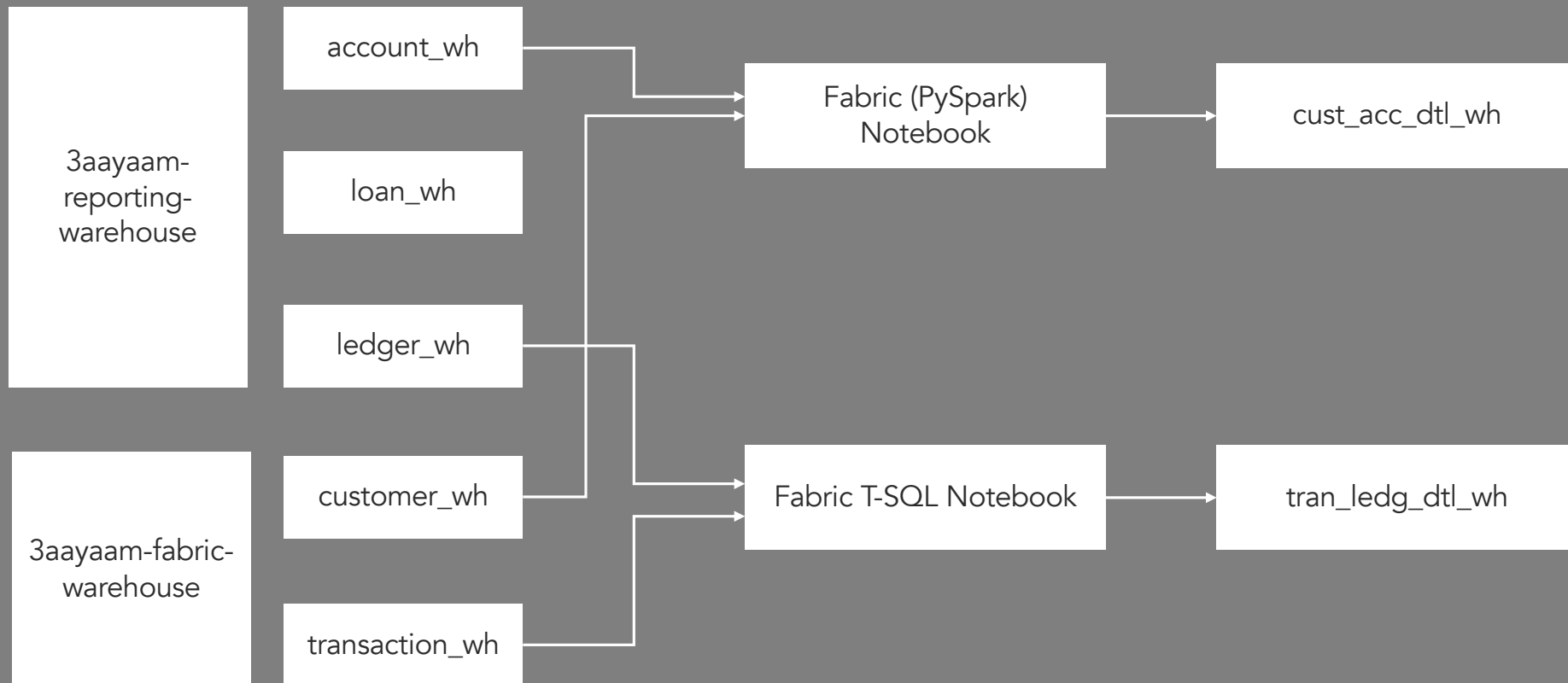


Multiple Warehouses

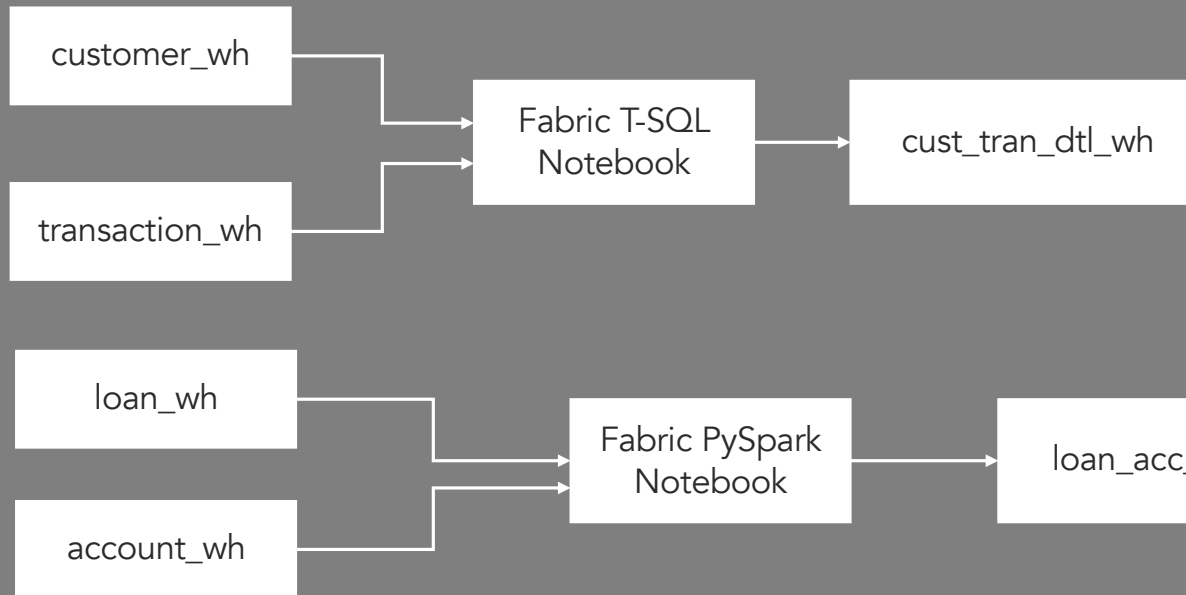


```
import com.microsoft.spark.fabric
from com.microsoft.spark.fabric.Constants import Constants
spark.read.synapsesql()
df.write.synapsesql()
```

Hands-On



Exercise – Warehouse



- Highest amount cross-country transaction should be loaded, for each country, along with customer details for source & target accounts.

- Load customer details with total no. of accounts, total account balance, total no. of loans and total loan amount. If total loan amount is less than 50% of total balance, then credit rating should be 9. If it is more than 50% and less than 60%, then credit rating should be 7. If it is more than 60% and less than 80%, then credit rating should be 6. If it is more than 80% and less than 100%, CR should be 5. If it is more than 100% CR should be 3

SQL

[Courses](#)[Blog](#)[About](#)[Contact](#)[FAQ](#)[Login](#)

SQL Essentials for Data Engineering



★★★★★ 5.0 (1 ratings)

Created by Soumyadeep Dey

English



SQL Essentials for Data Engineering

Buy now for FREE

Language: English

Instructors: Soumyadeep Dey

About the course

In this course you will get to learn about basic and advanced SQL that is needed for Data Engineering.

OLAP & Data Warehouse

[Courses](#)[Blog](#)[About](#)[Contact](#)[FAQ](#)[Login](#)

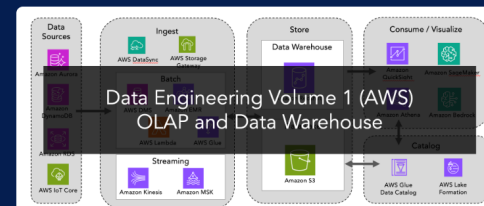
AWS OLAP & Data Warehouse - Data Engineering Vol 1

> Data Engineering Volume 1 (AWS)

★★★★★ 5.0 (1 ratings) Learners Enrolled : 22

Created by Soumyadeep Dey

English



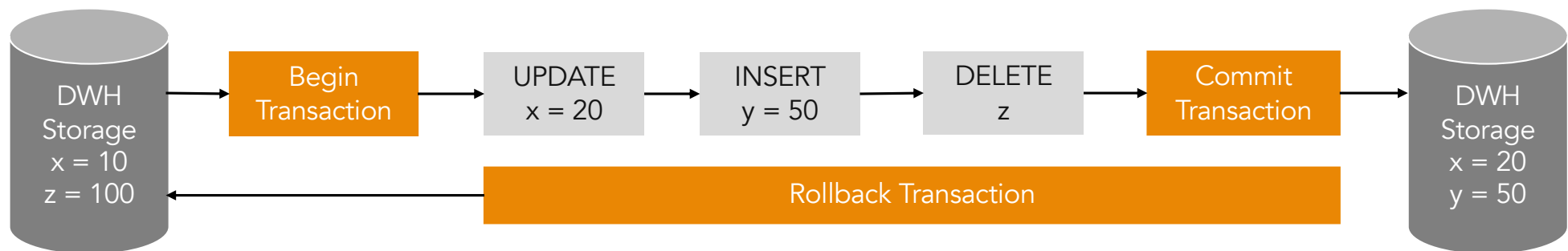
AWS OLAP & Data Warehouse - Data Engineering Vol 1

Buy now for ₹3,999

Language: English

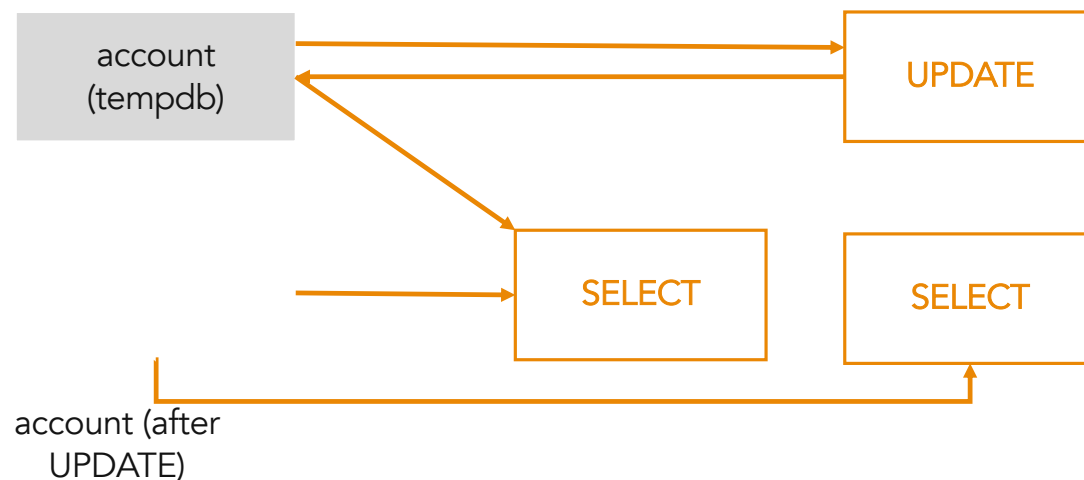
Transaction & Locking

- T-SQL (SQL Server 2008 & later versions).
- BEGIN TRANSACTION;
- COMMIT TRANSACTION;
- ROLLBACK TRANSACTION;
- Supports transactions across warehouses within the SAME workspace.



Transaction & Locking

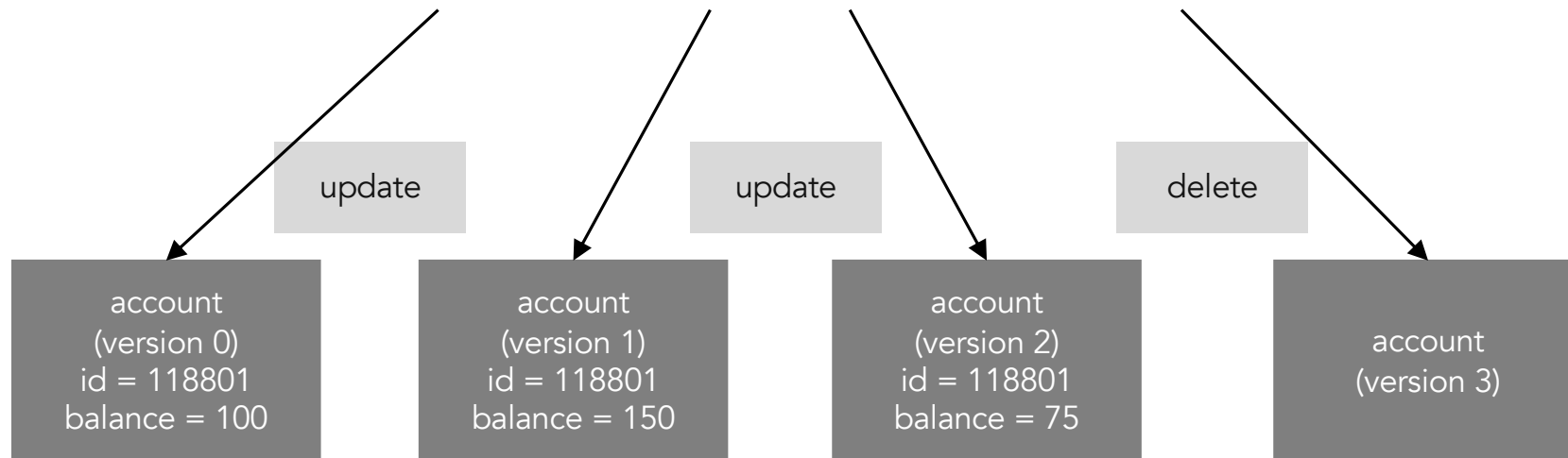
- Locks: Table level locking.
 - Snapshot Isolation – Row based isolation. MVCC using “tempdb”.
 - Schema Locks – Metadata locks.
 - Write-write conflict – Different row updates on the same table can conflict.



Time Travel Queries

- Query data in the past.
- Query earlier version of data. Max up to last 30 days.
- Syntax:

- **SELECT ... OPTION (FOR TIMESTAMP AS OF 'yyyy-mm-ddThh:mm:ss.fff')**



Data Clustering

Data clustering is used to organize and store data based on similarity. It improves query performance and reduces compute and storage access costs for queries by grouping similar records.

Source data

Sale ID	ProductID	SaleAmount	Region	Delivery_Date
42	770487	1172.51	North	2025-11-11
43	216739	3014.87	North	2025-11-29
44	126225	2811.49	North	2025-11-23
45	877572	3583.51	North	2025-12-12
46	388389	3510.21	North	2025-11-10
47	356787	2104.56	North	2025-11-09
...	...	...	...	...
1578	334053	2252.65	North	2025-12-11
1579	246316	1399.62	North	2025-11-08
1580	872246	4348.07	North	2025-12-18
1581	207473	3796.93	North	2026-01-10
1582	809570	808.38	North	2025-11-19
...	...	...	...	...
6947	876646	2120	North	2025-11-03
6948	671858	1398.02	North	2025-11-10
6949	191161	1085.99	North	2025-12-13
6950	719176	3820.31	North	2025-12-15
6951	542417	521.82	North	2025-12-22
6952	133326	1907.08	North	2025-11-10
6953	131244	1802.59	North	2026-01-07
6954	198246	1727.65	North	2026-02-06
6955	329258	1331.43	North	2025-12-03

Data Clustering
algorithm



Ordered table

File.parquet

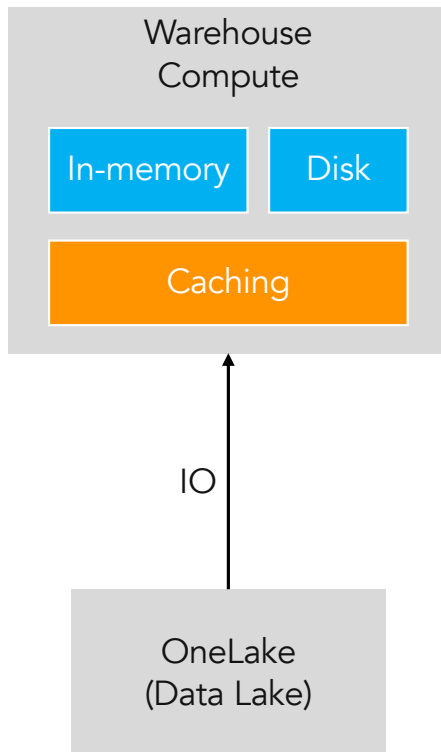
Sale ID	ProductID	SaleAmount	Region	Delivery_Date
6947	876646	2120	North	2025-11-03
1579	246316	1399.62	North	2025-11-08
47	356787	2104.56	North	2025-11-09
6948	671858	1398.02	North	2025-11-10
46	388389	3510.21	North	2025-11-10
6952	133326	1907.08	North	2025-11-10
42	770487	1172.51	North	2025-11-11
...	...	...	...	...
1582	809570	808.38	North	2025-11-19
44	126225	2811.49	North	2025-11-23
43	216739	3014.87	North	2025-11-29
6955	329258	1331.43	North	2025-12-03
...	...	...	...	...
1578	334053	2252.65	North	2025-12-11
45	877572	3583.51	North	2025-12-12
6949	191161	1085.99	North	2025-12-13
6950	719176	3820.31	North	2025-12-15
6951	542417	521.82	North	2025-12-22
...	...	...	...	...
6954	198246	1727.65	North	2026-02-06
6953	131244	1802.59	North	2026-01-07
1581	207473	3796.93	North	2026-01-10

Data Clustering

- Multiple queries with same columns in the WHERE clause.
- Large tables.
- Columns with high cardinality and WHERE clause in them.

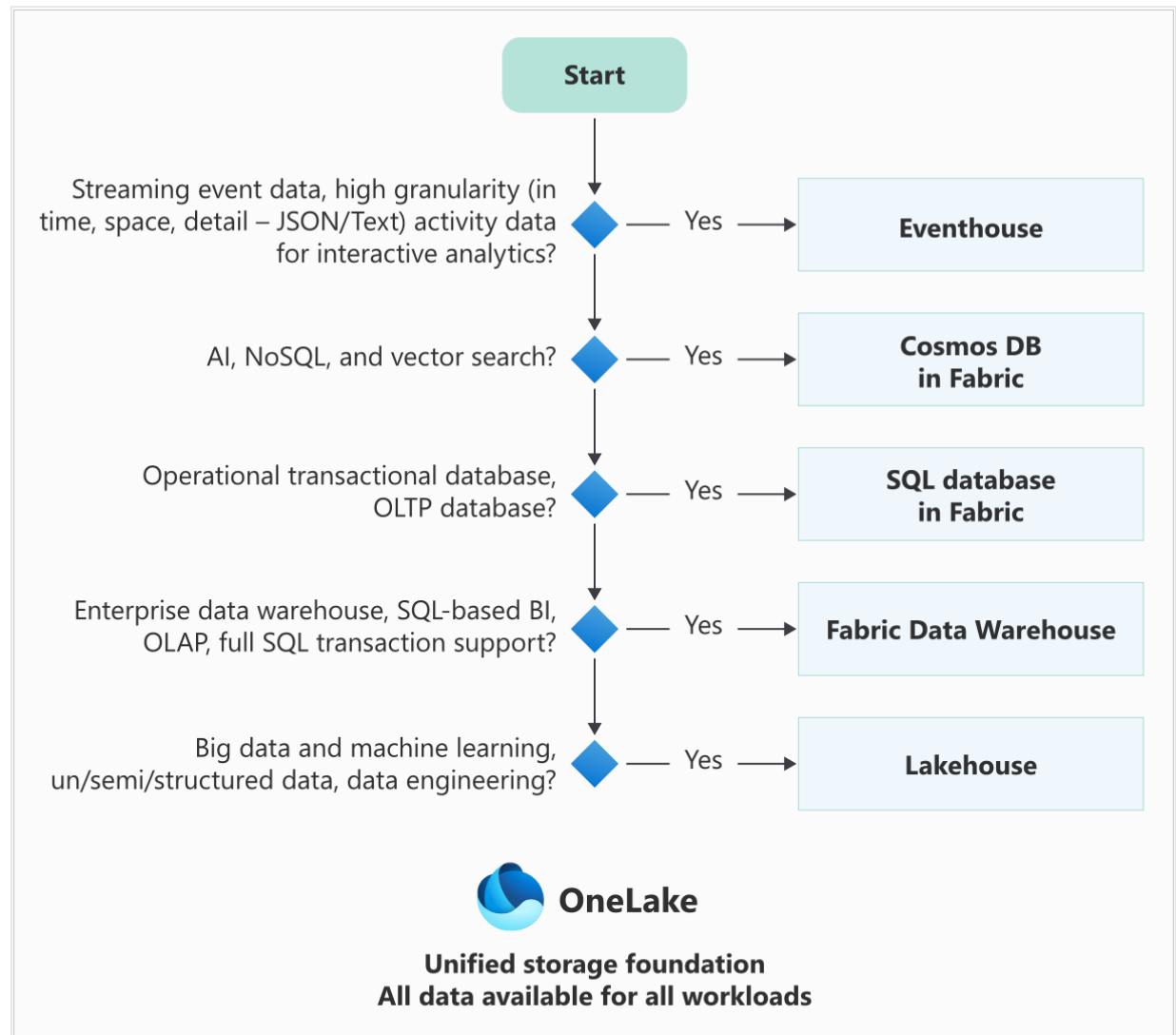
```
CREATE TABLE (  
    Column list  
)  
WITH (CLUSTER BY (col1, col2, ...));
```

Caching (Disk & Result Set)

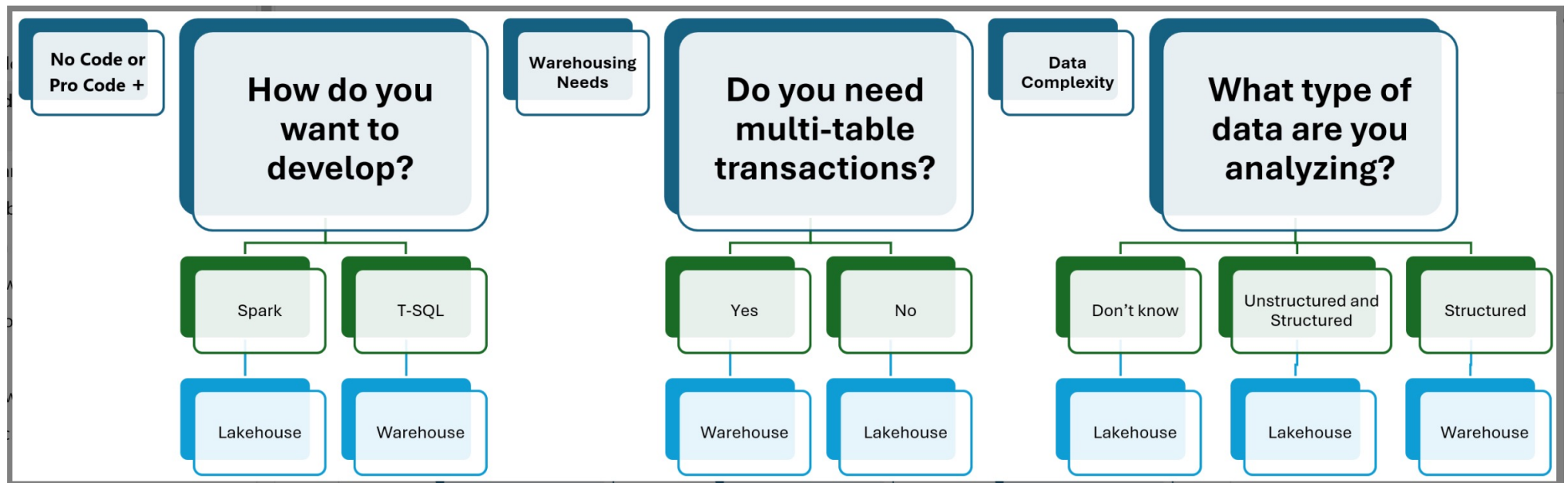


- Accessing rows from OneLake involves IO.
- Caching stores the data in the compute layer itself.
- In-memory Cache – Frequently accessed data is stored in memory. Columnar format.
- Disk Cache – Large datasets that cannot be stored in memory are stored in SSDs attached to Warehouse Compute.
- Managed by Fabric automatically.

Which Fabric Storage



Comparison: Lakehouse vs Warehouse



Summary

www.3aayaam.in

- Data Warehouse
- Loading DWH – ETL, ELT
- Dimensional Modeling
- Ingesting data into Fabric WH Tables
- Features:
 - Transaction, Locking
 - Clustering
 - Caching